

Project Vesta: Exploratory Operator–Spectral Analysis of Transformer Attention Dynamics

Introduction

This document summarizes an exploratory set of experiments studying the spectral behavior of transformer attention operators. The work began as a lightweight interpretability exercise rather than a formal attempt to derive a complete mathematical theory of transformer computation. The central question was whether attention operators exhibit recognizable spectral organization across layers, prompts, or semantic perturbations.

Motivation

Modern transformer models contain large collections of learned attention mechanisms whose internal organization remains difficult to interpret directly. Although attention maps alone do not fully explain model reasoning, they still provide a tractable object for analysis. The experiments in this project explored whether eigenvalue structure and spectral gap behavior might reveal useful organizational signatures inside transformer layers.

Rather than analyzing attention matrices in their raw asymmetric form, a symmetric surrogate operator was constructed:

$$H = 1/2(A + A^T)$$

This transformation makes the spectrum real-valued and allows conventional eigendecomposition methods to be applied more cleanly.

Experimental Setup

The experiments were performed using DistilGPT2 inside Google Colab. Attention matrices were extracted from multiple heads and layers. For each head:

- the attention matrix was extracted
- the symmetric surrogate operator was constructed
- eigenvalues were computed
- spectral gaps were measured

- layer-wise and head-wise comparisons were visualized

Prompt classes included arithmetic prompts, factual prompts, corrupted prompts, and nonsensical completions.

Preliminary Observations

Several recurring patterns emerged:

- spectral gaps varied across transformer layers
- later layers often exhibited stronger average spectral organization
- some heads appeared more perturbation-sensitive than others
- coherent and incoherent prompts produced localized spectral differences rather than complete collapse

Replacing semantically coherent completions such as “Paris” with corrupted alternatives such as “banana” did not destroy the spectral structure entirely. Instead, smaller redistributions appeared across specific heads and layers.

Limitations

The current framework has substantial limitations.

The symmetric operator construction discards directional information contained in the original attention matrix. Some observed structure may therefore be introduced by the symmetrization procedure itself.

In addition, the experiments were performed on a relatively small language model using short prompts. The present results are not sufficient to support strong claims about semantic representation or reasoning.

Why the Results Still Matter

Despite the limitations, the workflow itself proved reproducible and extensible.

The project demonstrated that:

- attention operators can be extracted systematically
- spectral diagnostics can be computed layer-by-layer
- perturbation studies can be visualized clearly
- experimental outputs remain stable enough to support iterative investigation

The project therefore moved beyond conceptual speculation into executable exploratory research.

Future Directions

Natural next steps include:

- testing larger transformer architectures
- studying non-symmetric operator formulations
- tracking spectral evolution during autoregressive generation
- clustering operators across semantic categories
- integrating spectral diagnostics with mechanistic interpretability workflows

Future work should prioritize reproducibility, sharper mathematical notation, and stronger statistical controls.